

BitRot Guard - Project Report & Viva Preparation

1. Project Overview & Features Implemented

BitRot Guard v3 is a modern web application designed to demonstrate the effects of silent data corruption ('bit rot') and how error-correcting codes, specifically the Hamming(7,4) algorithm, can detect and repair these errors automatically.

Key Features Implemented:

- Modern Web UI: Built with Flask, HTML/JS/CSS featuring a glassmorphism design.
- Hamming(7,4) Engine: Adds 3 parity bits for every 4 data bits, correcting single-bit errors per 7-bit block.
- Corruption Simulator: Injects random bit flips to simulate data degradation.
- Interactive Visualizations: Bit-level canvas grid, heatmap of corruption, and live integrity gauges.
- ECC Comparison Module: Evaluates 'No Protection', 'Simple Parity', and 'Hamming(7,4)'.
- AI Integration: Chatbot utilizing the Gemini API to explain concepts and analyze files.
- PDF Export: Generates a session report.

2. Detailed Implementation of Components

1. Backend (app.py): Flask server manages state and exposes RESTful API endpoints.
2. ECC Engine: Processes files byte-by-byte, calculates parity bits (XOR), and stores encoded data. During decoding, it calculates the syndrome to locate and flip back erroneous bits.
3. AI Module: Manages prompts to Google's Gemini API with fallback responses.
4. Frontend Visualizations: Uses HTML5 Canvas for the bit grid, Chart.js for the timeline, and SVG for gauges.

3. Possible Viva Questions & Answers

Q: What is bit rot and why is it a problem?

A: Bit rot occurs when data is silently corrupted over time, often due to hardware failures or storage degradation. It's a problem because the corruption is not immediately apparent, leading to data loss or system instability.

Q: How does the Hamming(7,4) code work?

A: It takes 4 data bits and adds 3 parity bits to create a 7-bit block. The parity bits are calculated using XOR operations on specific data bits. During decoding, the syndrome is calculated to identify and flip any erroneous bits.

Q: Why use Hamming(7,4) specifically?

A: It's a simple yet effective error-correcting code that can detect and correct single-bit errors. It's well-suited for educational purposes and small-scale data protection.

Q: What happens if two bits flip in the same 7-bit block?

A: The Hamming(7,4) code can only correct single-bit errors. If two bits flip, the syndrome will indicate an error, but the code will not be able to identify and flip the correct bits, leading to data corruption.

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Q: What is the purpose of Simple Parity in the comparison mode?

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Q: How is the web application structured?

A: Fla
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Q: What role does the AI Integration play?

A: It s
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